

[1]: # Purchase Pattern Analytics - Market Basket Analysis (Apriori)



[3]: `import sys`
`print(sys.executable)`

C:\ProgramData\anaconda3\python.exe

[4]: `import sys`
`!{sys.executable} -m pip install --user mysql-connector-python`

Requirement already satisfied: mysql-connector-python in c:\users\s\appdata\roaming\python\python313\site-packages (9.7.0)

[5]: `pip install mysql-connector-python`

Defaulting to user installation because normal site-packages is not writeable

Requirement already satisfied: mysql-connector-python in c:\users\s\appdata\roaming\python\python313\site-packages (9.7.0)

Note: you may need to restart the kernel to use updated packages.

[6]: `import mysql.connector`

[7]: `import sys`
`print(sys.executable)`

C:\ProgramData\anaconda3\python.exe

[8]: `!pip install mysql-connector-python`

Defaulting to user installation because normal site-packages is not writeable

Requirement already satisfied: mysql-connector-python in c:\users\s\appdata\roaming\python\python313\site-packages (9.7.0)

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```
[9]: import mysql.connector
```



```
print("MySQL Connector is working!")
```

MySQL Connector is working!

```
[10]: pip install flask
```

Defaulting to user installation because normal site-packages is not writeable

Requirement already satisfied: flask in c:\programdata\anaconda3\lib\site-packages (3.1.2)

Requirement already satisfied: blinker>=1.9.0 in c:\programdata\anaconda3\lib\site-packages (from flask) (1.9.0)

Requirement already satisfied: click>=8.1.3 in c:\programdata\anaconda3\lib\site-packages (from flask) (8.2.1)

Requirement already satisfied: itsdangerous>=2.2.0 in c:\programdata\anaconda3\lib\site-packages (from flask) (2.2.0)

Requirement already satisfied: jinja2>=3.1.2 in c:\programdata\anaconda3\lib\site-packages (from flask) (3.1.6)

Requirement already satisfied: markupsafe>=2.1.1 in c:\programdata\anaconda3\lib\site-packages (from flask) (3.0.2)

Requirement already satisfied: werkzeug>=3.1.0 in c:\programdata\anaconda3\lib\site-packages (from flask) (3.1.3)

Requirement already satisfied: colorama in c:\programdata\anaconda3\lib\site-packages (from click>=8.1.3->flask) (0.4.6)

Note: you may need to restart the kernel to use updated packages.

```
[11]: import mysql.connector
```

```
db = mysql.connector.connect(  
    host="18.136.157.135",  
    user="dm_team16",  
    password="2o_hihiFeTRE"  
)
```

```
cursor = db.cursor()  
cursor.execute("SHOW DATABASES")
```

```
for i in cursor:  
    print(i)
```

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```
('information_schema',)
('Automobile Insurance',)
('Fast food analysis',)
('OnlineBookstore',)
('US_Shoppers',)
('customer segmentation',)
('datamites_sql',)
('dell-company',)
('mysql',)
('onlinebookstore',)
('performance_schema',)
('project_automobile_insurance',)
('project_banking',)
('project_crop_production_india',)
('project_csat_analytics',)
('project_customer_segmentation',)
('project_food',)
('project_health_data',)
('project_hr_analytics',)
('project_itsm',)
('project_job_market_analysis',)
('project_medical_data_history',)
('project_movie_database',)
('project_orders',)
('project_profit_analysis',)
('project_purchase_pattern_analysis',)
('project_sales',)
('project_service_data',)
('project_shopping Data',)
('project_shopping_trends',)
('project_snack_analysis',)
('project_supply_chain_data',)
('project_telecom',)
('sys',)
('test',)
```

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```
[1]: import mysql.connector
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[13]: conn = mysql.connector.connect(
    host="18.136.157.135",
    user="dm_team16",
    password="2o_hihiFeTRE",
    database="project_purchase_pattern_analysis"
)

print("Database Connected Successfully")
```

Database Connected Successfully

```
[14]: cursor = conn.cursor()
cursor.execute("SHOW TABLES")

for table in cursor:
    print(table)

('mytable',)
```

```
[15]: df = pd.read_sql("SELECT * FROM mytable", conn)
df.head()
```

C:\Users\s\AppData\Local\Temp\ipykernel_1568\3142224628.py:1: UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or database string URI or sqlite3 DBAPI2 connection. Other DBAPI2 objects are not tested. Please consider using SQLAlchemy.

```
df = pd.read_sql("SELECT * FROM mytable", conn)
```

```
[15]:
```

	BillNo	Itemname	Quantity	Present_Date	Price	CustomerID	Country
0	536365	WHITE HANGING HEART T-LIGHT HOLDER	6	01-12-2010 08:26	2.55	17850	United Kingdom
1	536365	WHITE METAL LANTERN	6	01-12-2010 08:26	3.39	17850	United Kingdom
2	536365	CREAM CUPID HEARTS COAT HANGER	8	01-12-2010 08:26	2.75	17850	United Kingdom
3	536365	KNITTED UNION FLAG HOT WATER BOTTLE	6	01-12-2010 08:26	3.39	17850	United Kingdom
4	536365	RED WOOLLY HOTTIE WHITE HEART.	6	01-12-2010 08:26	3.39	17850	United Kingdom

```
[16]: df.columns
```

```
[16]: Index(['BillNo', 'Itemname', 'Quantity', 'Present_Date', 'Price', 'CustomerID',  
        'Country'],  
        dtype='object')
```

```
[17]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 522064 entries, 0 to 522063  
Data columns (total 7 columns):  
#   Column          Non-Null Count  Dtype  
---  ---  
0   BillNo          522064 non-null object  
1   Itemname        522064 non-null object  
2   Quantity        522064 non-null int64  
3   Present_Date    522064 non-null object  
4   Price           522064 non-null float64  
5   CustomerID      522064 non-null object  
6   Country         522064 non-null object  
dtypes: float64(1), int64(1), object(5)  
memory usage: 27.9+ MB
```

```
[18]: df = df.rename(columns={
    'BillNo': 'InvoiceNo',
    'Itemname': 'Description',
    'Quantity': 'Quantity'
})

df.head()
```

	InvoiceNo	Description	Quantity	Present_Date	Price	CustomerID	Country
0	536365	WHITE HANGING HEART T-LIGHT HOLDER	6	01-12-2010 08:26	2.55	17850	United Kingdom
1	536365	WHITE METAL LANTERN	6	01-12-2010 08:26	3.39	17850	United Kingdom
2	536365	CREAM CUPID HEARTS COAT HANGER	8	01-12-2010 08:26	2.75	17850	United Kingdom
3	536365	KNITTED UNION FLAG HOT WATER BOTTLE	6	01-12-2010 08:26	3.39	17850	United Kingdom
4	536365	RED WOOLLY HOTTIE WHITE HEART.	6	01-12-2010 08:26	3.39	17850	United Kingdom

```
[19]: # Remove missing Invoice or Product names
df = df.dropna(subset=['InvoiceNo', 'Description'])

# Convert InvoiceNo to string
df['InvoiceNo'] = df['InvoiceNo'].astype(str)

# Remove cancelled invoices (if any)
df = df[~df['InvoiceNo'].str.startswith('C')]

# Keep only positive quantity
df['Quantity'] = pd.to_numeric(df['Quantity'], errors='coerce')
df = df[df['Quantity'] > 0]

df.head()
```

[19]:	InvoiceNo	Description	Quantity	Present_Date	Price	CustomerID	Country
0	536365	WHITE HANGING HEART T-LIGHT HOLDER	6	01-12-2010 08:26	2.55	17850	United Kingdom
1	536365	WHITE METAL LANTERN	6	01-12-2010 08:26	3.39	17850	United Kingdom
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4	536365	RED WOOLLY HOTTIE WHITE HEART.	6	01-12-2010 08:26	3.39	17850	United Kingdom

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```
[20]: transactions = (
    df.groupby('InvoiceNo')['Description']
      .apply(list)
      .tolist()
)

transactions[:5]
```

```
[20]: [['WHITE HANGING HEART T-LIGHT HOLDER',
        'WHITE METAL LANTERN',
        'CREAM CUPID HEARTS COAT HANGER',
        'KNITTED UNION FLAG HOT WATER BOTTLE',
        'RED WOOLLY HOTTIE WHITE HEART.',
        'SET 7 BABUSHKA NESTING BOXES',
        'GLASS STAR FROSTED T-LIGHT HOLDER'],
        ['HAND WARMER UNION JACK', 'HAND WARMER RED POLKA DOT'],
        ['ASSORTED COLOUR BIRD ORNAMENT',
        'POPPY'S PLAYHOUSE BEDROOM',
        'POPPY'S PLAYHOUSE KITCHEN',
        'FELTCRAFT PRINCESS CHARLOTTE DOLL',
        'IVORY KNITTED MUG COSY',
        'BOX OF 6 ASSORTED COLOUR TEASPOONS',
        'BOX OF VINTAGE JIGSAW BLOCKS',
        'BOX OF VINTAGE ALPHABET BLOCKS',
        'HOME BUILDING BLOCK WORD',
        'LOVE BUILDING BLOCK WORD',
        'RECIPE BOX WITH METAL HEART',
        'DOORMAT NEW ENGLAND'],
        ['JAM MAKING SET WITH JARS',
        'RED COAT RACK PARIS FASHION',
        'YELLOW COAT RACK PARIS FASHION',
        'BLUE COAT RACK PARIS FASHION'],
        ['BATH BUILDING BLOCK WORD']]
```

```
[24]: !pip install mlxtend
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: mlxtend in c:\users\s\appdata\roaming\python\python313\site-packages (0.25.0)
Requirement already satisfied: scipy>=1.16.3 in c:\programdata\anaconda3\lib\site-packages (from mlxtend) (1.16.3)
Requirement already satisfied: numpy>=2.3.5 in c:\programdata\anaconda3\lib\site-packages (from mlxtend) (2.3.5)
Requirement already satisfied: pandas>=2.3.3 in c:\programdata\anaconda3\lib\site-packages (from mlxtend) (2.3.3)
Requirement already satisfied: scikit-learn>=1.8.0 in c:\users\s\appdata\roaming\python\python313\site-packages (from mlxtend) (1.9.0)
Requirement already satisfied: matplotlib>=3.10.8 in c:\users\s\appdata\roaming\python\python313\site-packages (from mlxtend) (3.11.0)
Requirement already satisfied: joblib>=1.5.2 in c:\programdata\anaconda3\lib\site-packages (from mlxtend) (1.5.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\programdata\anaconda3\lib\site-packages (from matplotlib>=3.10.8->mlxtend) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\programdata\anaconda3\lib\site-packages (from matplotlib>=3.10.8->mlxtend) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\programdata\anaconda3\lib\site-packages (from matplotlib>=3.10.8->mlxtend) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\programdata\anaconda3\lib\site-packages (from matplotlib>=3.10.8->mlxtend) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\programdata\anaconda3\lib\site-packages (from matplotlib>=3.10.8->mlxtend) (25.0)
Requirement already satisfied: pillow>=9 in c:\programdata\anaconda3\lib\site-packages (from matplotlib>=3.10.8->mlxtend) (12.0.0)
Requirement already satisfied: pyparsing>=3 in c:\programdata\anaconda3\lib\site-packages (from matplotlib>=3.10.8->mlxtend) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in c:\programdata\anaconda3\lib\site-packages (from matplotlib>=3.10.8->mlxtend) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\programdata\anaconda3\lib\site-packages (from pandas>=2.3.3->mlxtend) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\programdata\anaconda3\lib\site-packages (from pandas>=2.3.3->mlxtend) (2025.2)
Requirement already satisfied: six>=1.5 in c:\programdata\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib>=3.10.8->mlxtend) (1.17.0)
Requirement already satisfied: narwhals>=2.0.1 in c:\programdata\anaconda3\lib\site-packages (from scikit-learn>=1.8.0->mlxtend) (2.7.0)
Requirement already satisfied: threadpoolctl>=3.5.0 in c:\programdata\anaconda3\lib\site-packages (from scikit-learn>=1.8.0->mlxtend) (3.5.0)
```

```
[25]: from mlxtend.preprocessing import TransactionEncoder
```

```
[26]: # one-hot encoding
```

```
te = TransactionEncoder()
te_array = te.fit(transactions).transform(transactions)
basket = pd.DataFrame(te_array, columns=te.columns_)

basket.head()
```

[26]:

		*Boombox Ipod Classic	*USB Office Mirror Ball	10 COLOUR SPACEBOY PEN	12 COLOURED PARTY BALLOONS	12 DAISY PEGS IN WOOD BOX	12 EGG HOUSE PAINTED WOOD	12 HANGING EGGS HAND PAINTED	12 IVORY ROSE PEG PLACE SETTINGS	12 MESSAGE CARDS WITH ENVELOPES	...	returned	taig adjust	test	to push order througha s stock was	website fixed	wrongly coded 20713
0	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False	False	False

5 rows × 4057 columns



[27]:

```
# Count how many times each product appears
item_freq = basket.sum()

# Keep only products bought at Least 100 times
basket_reduced = basket[item_freq[item_freq >= 100].index]

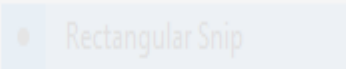
basket_reduced.shape
```

[27]: (20327, 1532)

[28]: `from mlxtend.frequent_patterns import apriori`

```
frequent_itemsets = apriori(
    basket_reduced,
    min_support=0.02, # increase support
    use_colnames=True
)

frequent_itemsets.head()
```



[28]:

	support	itemsets
0	0.029124	()
1	0.022482	(3 STRIPEY MICE FELTCRAFT)
2	0.022827	(4 TRADITIONAL SPINNING TOPS)
3	0.046342	(6 RIBBONS RUSTIC CHARM)
4	0.020515	(60 CAKE CASES DOLLY GIRL DESIGN)

	support	itemsets
0	0.029124	()
1	0.022482	(3 STRIPEY MICE FELTCRAFT)
2	0.022827	(4 TRADITIONAL SPINNING TOPS)
3	0.046342	(6 RIBBONS RUSTIC CHARM)
4	0.020515	(60 CAKE CASES DOLLY GIRL DESIGN)

[29]: `frequent_itemsets.sort_values(by='support', ascending=False).head(10)`

[29]:

	support	itemsets
263	0.108329	(WHITE HANGING HEART T-LIGHT HOLDER)
98	0.101540	(JUMBO BAG RED RETROSPOT)
191	0.093669	(REGENCY CAKESTAND 3 TIER)
152	0.081468	(PARTY BUNTING)
123	0.075810	(LUNCH BAG RED RETROSPOT)
16	0.070399	(ASSORTED COLOUR BIRD ORNAMENT)
210	0.066217	(SET OF 3 CAKE TINS PANTRY DESIGN)
144	0.062921	(PACK OF 72 RETROSPOT CAKE CASES)
116	0.061987	(LUNCH BAG BLACK SKULL)
134	0.060609	(NATURAL SLATE HEART CHALKBOARD)

	support	itemsets
263	0.108329	(WHITE HANGING HEART T-LIGHT HOLDER)
98	0.101540	(JUMBO BAG RED RETROSPOT)
191	0.093669	(REGENCY CAKESTAND 3 TIER)
152	0.081468	(PARTY BUNTING)
123	0.075810	(LUNCH BAG RED RETROSPOT)
16	0.070399	(ASSORTED COLOUR BIRD ORNAMENT)
210	0.066217	(SET OF 3 CAKE TINS PANTRY DESIGN)
144	0.062921	(PACK OF 72 RETROSPOT CAKE CASES)
116	0.061987	(LUNCH BAG BLACK SKULL)
134	0.060609	(NATURAL SLATE HEART CHALKBOARD)

```
[30]: from mlxtend.frequent_patterns import association_rules

rules = association_rules(
    frequent_itemsets,
    metric='lift',
    min_threshold=1
)

rules[['antecedents', 'consequents', 'support', 'confidence', 'lift']].head()
```

[30]:

	antecedents	consequents	support	confidence	lift
0	(PACK OF 72 RETROSPOT CAKE CASES)	(60 TEATIME FAIRY CAKE CASES)	0.021400	0.340109	8.663415
1	(60 TEATIME FAIRY CAKE CASES)	(PACK OF 72 RETROSPOT CAKE CASES)	0.021400	0.545113	8.663415
2	(ALARM CLOCK BAKELIKE PINK)	(ALARM CLOCK BAKELIKE GREEN)	0.020219	0.541502	11.382741
3	(ALARM CLOCK BAKELIKE GREEN)	(ALARM CLOCK BAKELIKE PINK)	0.020219	0.425026	11.382741
4	(ALARM CLOCK BAKELIKE RED)	(ALARM CLOCK BAKELIKE GREEN)	0.031141	0.616358	12.956273

```
[31]: strong_rules = rules[
    (rules['confidence'] >= 0.6) &
    (rules['lift'] > 1)
]

strong_rules
```

[31]:

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty
4	(ALARM CLOCK BAKELIKE RED)	(ALARM CLOCK BAKELIKE GREEN)	0.050524	0.047572	0.031141	0.616358	12.956273	1.0	0.028737	2.482597	0.971923	0.465099	0.597196
5	(ALARM CLOCK BAKELIKE GREEN)	(ALARM CLOCK BAKELIKE RED)	0.047572	0.050524	0.031141	0.654602	12.956273	1.0	0.028737	2.748932	0.968911	0.465099	0.636222

11	(CHARLOTTE BAG PINK POLKADOT)	(RED RETROSPOT CHARLOTTE BAG)	0.036306	0.050573	0.025582	0.704607	13.932439	1.0	0.023746	3.214115	0.963195	0.417335	0.688872
18	(DOLLY GIRL LUNCH BOX)	(SPACEBOY LUNCH BOX)	0.040832	0.043489	0.025828	0.632530	14.544615	1.0	0.024052	2.602964	0.970890	0.441548	0.615823
20	(DOTCOM POSTAGE)	(JUMBO BAG RED RETROSPOT)	0.034831	0.101540	0.023860	0.685028	6.746400	1.0	0.020323	2.852510	0.882511	0.212068	0.649432
22	(GARDENERS KNEELING PAD KEEP CALM)	(GARDENERS KNEELING PAD CUP OF TEA)	0.044522	0.037094	0.026762	0.601105	16.205120	1.0	0.025111	2.413935	0.982012	0.487892	0.585739
23	(GARDENERS KNEELING PAD CUP OF TEA)	(GARDENERS KNEELING PAD KEEP CALM)	0.037094	0.044522	0.026762	0.721485	16.205120	1.0	0.025111	3.430621	0.974436	0.487892	0.708508
24	(GREEN REGENCY TEACUP AND SAUCER)	(PINK REGENCY TEACUP AND SAUCER)	0.047966	0.036208	0.029763	0.620513	17.137451	1.0	0.028027	2.539722	0.989091	0.547016	0.606256
25	(PINK REGENCY TEACUP AND SAUCER)	(GREEN REGENCY TEACUP AND SAUCER)	0.036208	0.047966	0.029763	0.822011	17.137451	1.0	0.028027	5.348834	0.977024	0.547016	0.813043
28	(ROSES REGENCY TEACUP AND SAUCER)	(GREEN REGENCY TEACUP AND SAUCER)	0.049835	0.047966	0.035962	0.721619	15.044460	1.0	0.033572	3.419896	0.982493	0.581543	0.707593
29	(GREEN REGENCY TEACUP AND SAUCER)	(ROSES REGENCY TEACUP AND SAUCER)	0.047966	0.049835	0.035962	0.749744	15.044460	1.0	0.033572	3.796765	0.980564	0.581543	0.736618

36	(JUMBO BAG BAROQUE BLACK WHITE)	(JUMBO BAG RED RETROSPOT)	0.045506	0.101540	0.028533	0.627027	6.175183	1.0	0.023913	2.408915	0.878016	0.240764	0.584875
47	(JUMBO BAG PINK POLKADOT)	(JUMBO BAG RED RETROSPOT)	0.059576	0.101540	0.040291	0.676301	6.660447	1.0	0.034242	2.775600	0.903699	0.333469	0.639718
59	(JUMBO BAG SCANDINAVIAN BLUE PAISLEY)	(JUMBO BAG RED RETROSPOT)	0.032272	0.101540	0.021794	0.675305	6.650641	1.0	0.018517	2.767089	0.877973	0.194554	0.638609
60	(JUMBO BAG SPACEBOY DESIGN)	(JUMBO BAG RED RETROSPOT)	0.033551	0.101540	0.020957	0.624633	6.151610	1.0	0.017551	2.393554	0.866514	0.183621	0.582211
62	(JUMBO BAG STRAWBERRY)	(JUMBO BAG RED RETROSPOT)	0.040094	0.101540	0.026024	0.649080	6.392366	1.0	0.021953	2.560297	0.878798	0.225106	0.609420
66	(JUMBO BAG WOODLAND ANIMALS)	(JUMBO BAG RED RETROSPOT)	0.042013	0.101540	0.026221	0.624122	6.146571	1.0	0.021955	2.390296	0.874028	0.223480	0.581642
73	(JUMBO STORAGE BAG SUKI)	(JUMBO BAG RED RETROSPOT)	0.057952	0.101540	0.035470	0.612054	6.027727	1.0	0.029586	2.315943	0.885412	0.285998	0.568210
134	(PAPER CHAIN KIT VINTAGE CHRISTMAS)	(PAPER CHAIN KIT 50'S CHRISTMAS)	0.039898	0.056231	0.026812	0.672010	11.950958	1.0	0.024568	2.877432	0.954403	0.386799	0.652468
139	(PINK REGENCY TEACUP AND SAUCER)	(ROSES REGENCY TEACUP AND SAUCER)	0.036208	0.049835	0.028042	0.774457	15.540353	1.0	0.026237	4.212779	0.970802	0.483461	0.762627
140	(RED HANGING HEART T-LIGHT HOLDER)	(WHITE HANGING HEART T- LIGHT HOLDER)	0.035470	0.108329	0.023565	0.664355	6.132764	1.0	0.019722	2.656591	0.867719	0.195990	0.623578

143	(STRAWBERRY CHARLOTTE BAG)	(RED RETROSPOT CHARLOTTE BAG)	0.035224	0.050573	0.023712	0.673184	13.311107	1.0	0.021931	2.905084	0.958642	0.381933	0.655776
145	(WOODLAND CHARLOTTE BAG)	(RED RETROSPOT CHARLOTTE BAG)	0.040685	0.050573	0.024745	0.608222	12.026594	1.0	0.022688	2.423383	0.955735	0.372041	0.587354
150	(ROSES REGENCY TEACUP AND SAUCER, GREEN REGENC...	(PINK REGENCY TEACUP AND SAUCER)	0.035962	0.036208	0.025336	0.704514	19.457423	1.0	0.024034	3.261722	0.983992	0.540966	0.693413
151	(PINK REGENCY TEACUP AND SAUCER, GREEN REGENCY...	(ROSES REGENCY TEACUP AND SAUCER)	0.029763	0.049835	0.025336	0.851240	17.081095	1.0	0.023852	6.387219	0.970336	0.466908	0.843437
152	(ROSES REGENCY TEACUP AND SAUCER, PINK REGENCY...	(GREEN REGENCY TEACUP AND SAUCER)	0.028042	0.047966	0.025336	0.903509	18.836536	1.0	0.023991	9.866537	0.974231	0.500000	0.898647
155	(PINK REGENCY TEACUP AND SAUCER)	(ROSES REGENCY TEACUP AND SAUCER, GREEN REGENC...	0.036208	0.035962	0.025336	0.699728	19.457423	1.0	0.024034	3.210552	0.984243	0.540966	0.688527
158	(JUMBO STORAGE BAG SUKI, JUMBO BAG PINK POLKADOT)	(JUMBO BAG RED RETROSPOT)	0.025287	0.101540	0.020269	0.801556	7.894010	1.0	0.017701	4.527535	0.895978	0.190212	0.779129

```
[33]: strong_rules.to_csv("apriori_results.csv", index=False)
print("Results Saved Successfully")
```

Results Saved Successfully

```
[34]: # Top 10 Frequent itemsets
frequent_itemsets.sort_values(by='support', ascending=False).head(10)
```

[34]:	support	itemsets
263	0.108329	(WHITE HANGING HEART T-LIGHT HOLDER)
98	0.101540	(JUMBO BAG RED RETROSPOT)
191	0.093669	(REGENCY CAKESTAND 3 TIER)
152	0.081468	(PARTY BUNTING)
123	0.075810	(LUNCH BAG RED RETROSPOT)
16	0.070399	(ASSORTED COLOUR BIRD ORNAMENT)
210	0.066217	(SET OF 3 CAKE TINS PANTRY DESIGN)
144	0.062921	(PACK OF 72 RETROSPOT CAKE CASES)
116	0.061987	(LUNCH BAG BLACK SKULL)
134	0.060609	(NATURAL SLATE HEART CHALKBOARD)

```
[35]: strong_rules[['antecedents', 'consequents', 'support', 'confidence', 'lift']].sort_values(by='lift', ascending=False).head(10)
```

[35]:	antecedents	consequents	support	confidence	lift
150	(ROSES REGENCY TEACUP AND SAUCER, GREEN REGENC...	(PINK REGENCY TEACUP AND SAUCER)	0.025336	0.704514	19.457423
155	(PINK REGENCY TEACUP AND SAUCER)	(ROSES REGENCY TEACUP AND SAUCER, GREEN REGENC...	0.025336	0.699728	19.457423
152	(ROSES REGENCY TEACUP AND SAUCER, PINK REGENCY...	(GREEN REGENCY TEACUP AND SAUCER)	0.025336	0.903509	18.836536
24	(GREEN REGENCY TEACUP AND SAUCER)	(PINK REGENCY TEACUP AND SAUCER)	0.029763	0.620513	17.137451
25	(PINK REGENCY TEACUP AND SAUCER)	(GREEN REGENCY TEACUP AND SAUCER)	0.029763	0.822011	17.137451
151	(PINK REGENCY TEACUP AND SAUCER, GREEN REGENCY...	(ROSES REGENCY TEACUP AND SAUCER)	0.025336	0.851240	17.081095
22	(GARDENERS KNEELING PAD KEEP CALM)	(GARDENERS KNEELING PAD CUP OF TEA)	0.026762	0.601105	16.205120
23	(GARDENERS KNEELING PAD CUP OF TEA)	(GARDENERS KNEELING PAD KEEP CALM)	0.026762	0.721485	16.205120
139	(PINK REGENCY TEACUP AND SAUCER)	(ROSES REGENCY TEACUP AND SAUCER)	0.028042	0.774457	15.540353
29	(GREEN REGENCY TEACUP AND SAUCER)	(ROSES REGENCY TEACUP AND SAUCER)	0.035962	0.749744	15.044460